Midterm 2 Part a

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I. Questions (39 points)

1. How many input and output units (total) are there for a linear perceptron with d = 5 for binary classification?

a) 4
b) 5
c) 6
d) 7
e) 8

2. How many weights are there for a linear perceptron with d = 5 for binary classification?

a) 4
b) 5
c) 6
d) 7
e) 8

3. How many input and output units (total) are there for a linear perceptron with d = 5 for K = 4 classification?

a) 8
b) 9
c) 10
d) 15
e) 24

4. How many weights are there for a linear perceptron with d = 5 for K = 4 classification?

a) 8
b) 9
c) 10
d) 15
e) 24

5. Which of the following functions can not be learned by linear perceptron?

a) and
b) or
c) xor
d) not

6. How many input, hidden, output units (total) are there for a multi layer perceptron with d = 5, hidden node count 3, for binary classification?

a) 9
b) 12
c) 16
d) 22
e) 32

7. How many weights are there for a multi layer perceptron with d = 5, hidden node count 3, for binary classification?

a) 9
b) 12
c) 16
d) 22
e) 32

8. How many input and output units (total) are there for a multilayer perceptron with d = 5, hidden node count 3, for K = 4 classification?

a) 10
b) 14
c) 20
d) 32
e) 34

9. How many weights are there for a multilayer perceptron with d = 5, hidden node count 3, for K = 4 classification?

a) 10
b) 14
c) 20
d) 32
e) 34

10. Which type of function will not result in a nonlinear function when used as a basis function in linear discrimination?

a) sin
b) exp
c) linear
d) log

11. How many separate hyperplanes (discriminant functions) are defined when K = 10 in pairwise linear classification?

a) 2
b) 5
c) 9
d) 10
e) 45

12. Which of the following is the sigmoid function?

a) $\frac{1}{x}$
b) $\frac{1}{lnx}$
c) $\frac{1}{1+lnx}$
d) $\frac{1}{1+e^{-x}}$
e) $\frac{1}{1+e^{x}}$

13. What is the value of sigmoid function when x = ∞ and x = -∞?

a) 1, 0
b) 0, 1
c) -1, 1
d) 1, -1
e) -1, 0

14. What is the derivative of the sigmoid function y?

a) y
b) 1 - y
c) y (1 - y)
d) ln y
e) $\frac{1}{y}$

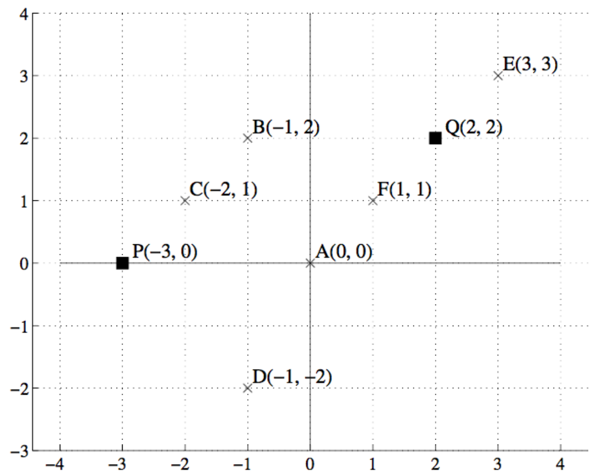
15. Which of the following is the softmax function?

a) $\frac{1}{x}$
b) $\frac{e^x}{\sum_j^K e^j}$
c) $\frac{1}{1+lnx}$
d) $\frac{1}{1+e^{-x}}$
e) $\frac{1}{1+e^{x}}$

16. If there are 3 classes, which of the following vector can be the output of the softmax function?

a) [0.3 0.3 0.3]
b) [0.3 0.3 0.4]
c) [0.4 0.4 0.4]
d) [1.0 1.0 1.0]
e) [0.5 0.5 0.5]

Suppose we are given the following dataset. In the following questions 51-54, we will do k-means clustering to cluster the points A, B ... F into 2 clusters. The current cluster centers are P and Q.



17. Which points are assigned to cluster center P?

a) A, B, C, D
b) B, C, D
c) B, C
d) A, B, C, D, F
e) C

18. Which points are assigned to cluster center Q?

a) E, F
b) E
c) A, F, E
d) A, B, F, E
e) F

19. What does cluster center P get updated to?

a) -1, 1/4
b) -4/3, 1 / 3
c) -3 / 2, 3 / 2
d) -3 / 5, 2 / 5
e) -2, 1

20. What does cluster center Q get updated to?

a) 2, 2
b) 3, 3
c) 4 / 3, 4 / 3
d) 3 / 4, 6 / 4
e) 1, 1

21. Which of the following is a feature extraction technique?

a) Forward selection
b) Backward selection
c) Floating selection
d) PCA

22. If the eigenvalues of the covariance matrix are λ_1 = 2, λ_2 = 1, λ_3 = 0.5, λ_4 = 0, and λ_5 = 0; then with proportion of variance explained 90 percent, what will be the reduced dimensionality?

a) 1
b) 2
c) 3
d) 4
e) 5

23. Which of the following a subset selection technique starts with empty subset of features and adds features one by one?

a) Forward selection
b) Backward selection
c) Floating selection
d) Exhaustive search

24. Which of the following a subset selection technique starts with full set of features and removes features one by one?

a) Forward selection
b) Backward selection
c) Floating selection
d) Exhaustive search

25. What is the number of possible subsets of d variables?

a) d
b) d^2
c) 2^d
d) d^3

26. No free ... theorem that says there is no such thing as the best learning algorithm?

a) Bed
b) Lunch
c) Home
d) Brunch